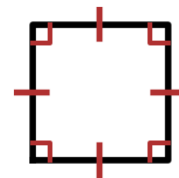
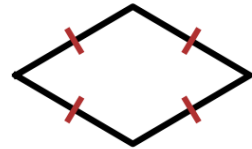


Speak Like A Mathematician:

rectangle:

rhombus:

square:



Special Parallelogram Theorems

Every rectangle, rhombus, and square is a _____, so each inherits all four parallelogram properties.

A parallelogram is a rectangle if and only if its diagonals are _____.

A parallelogram is a rhombus if and only if its diagonals are _____ — and each diagonal bisects a pair of opposite _____.

Example 1 - Check Every Property

Mark **X** for every property that is ALWAYS true of the figure. A property must hold for EVERY example of the shape to earn its **X**.

Property		Rect.	Rhom.	Sq.
Both pairs of opposite sides $\parallel$				
Both pairs of opposite sides $\cong$				
Both pairs of opposite angles $\cong$				
Diagonals bisect each other				
Four right angles				
All four sides $\cong$				
Diagonals are congruent				
Diagonals are perpendicular				
Each diagonal bisects opposite $\angle$ s				

a) Why does every column get the first four **X**s?

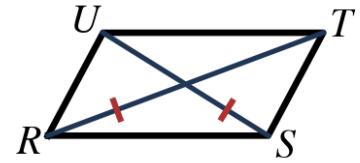
b) Which shape earns every row? Why?

Example 2 - Congruent Diagonals Make a Rectangle

Given: RSTU is a parallelogram with $\overline{RT} \cong \overline{SU}$

Prove: RSTU is a rectangle.

Statements	Reasons
1. RSTU is a $\square$ with $\overline{RT} \cong \overline{SU}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.

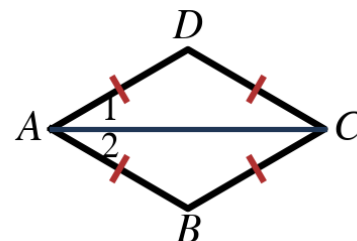


Example 3 - A Rhombus Diagonal Bisects the Angles

Given: ABCD is a rhombus with diagonal $\overline{AC}$.

Prove: $\angle 1 \cong \angle 2$ ($\overline{AC}$ bisects $\angle BAD$)

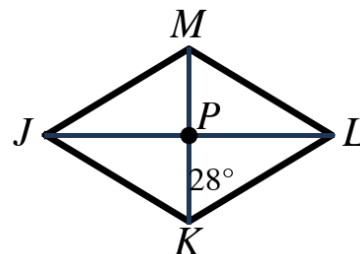
Statements	Reasons
1. ABCD is a rhombus with diagonal $\overline{AC}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.



Example 4 - Rhombus Numbers

In rhombus JKLM at the right, the diagonals meet at P and $m\angle MKL = 28^\circ$, $JP = 9$.

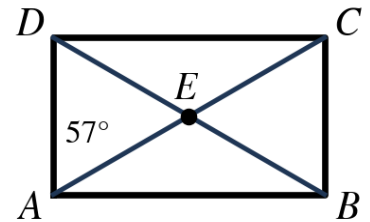
- Find $m\angle JKM$ and $m\angle JKL$.
- Find $m\angle JPK$.
- Find JL and $m\angle KLM$.



Example 5 - Rectangle Numbers

In rectangle ABCD at the right, the diagonals meet at E, $m\angle CAD = 57^\circ$, and $BD = 26$.

- Find $m\angle BAC$.
- Find AC and AE.
- Find $m\angle AEB$.



Example 6 - Square Numbers

In square PQRS at the right, the diagonals meet at X and $QX = 11$.

- Find $m\angle PXQ$ and $m\angle RPS$.
- Find PR.
- Find PQ. Leave your answer exact.

